R-INVARIANT LOCALLY CLOSED SUBSPACES GIVE IMMERSIONS OF QUOTIENT STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1935-1948 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 1950-1980, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_stacks-properties.json (pb=1120, stmt 1935-1948, proof 1950-1980).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04TK: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1980–1980. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-check-property-covering: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 683–693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-check-representable-covering: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 394–401. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 044F: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1980–1980. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04ZN: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1980–1980. Cited by the retained statement or

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let (U, R, s, t, c) be a smooth groupoid in algebraic spaces. Let $\mathcal{X} = [U/R]$ be the associated algebraic stack, see Algebraic Stacks, Theorem 04TK. Let $Z \subset U$ be an R-invariant locally closed subspace. Then*

$$[Z/R_Z] \longrightarrow [U/R]$$

is an immersion of algebraic stacks, where R_Z is the restriction of R to Z. If $Z \subset U$ is open (resp. closed) then the morphism is an open (resp. closed) immersion of algebraic stacks.

Proof. Recall that by Groupoids in Spaces, Definition 044F (see also discussion following the definition) we have $R_Z = s^{-1}(Z) = t^{-1}(Z)$ as locally closed subspaces of R. Hence the two morphisms $R_Z \to Z$ are smooth as base changes of s and t. Hence $(Z, R_Z, s|_{R_Z}, t|_{R_Z}, c|_{R_Z \times_{s,Z,t} R_Z})$ is a smooth groupoid in algebraic spaces, and we see that $[Z/R_Z]$ is an algebraic stack, see Algebraic Stacks, Theorem 04TK. The assumptions of Groupoids in Spaces, Lemma 04ZN are all satisfied and it follows that we have a 2-fibre square

$$\begin{array}{ccc} Z & \longrightarrow & [Z/R_Z] \\ \downarrow & & \downarrow \\ U & \longrightarrow & [U/R] \end{array}$$

It follows from this and Lemma lemma-check-representable-covering that $[Z/R_Z] \to [U/R]$ is representable by algebraic spaces, whereupon it follows from Lemma lemma-check-property-covering that the right vertical arrow is an immersion (resp. closed immersion, resp. open immersion) if and only if the left vertical arrow is. $\square$